

4. KV4 - Izrada konačne vizualizacije podataka

4.1. Implementacija osnovnih funkcionalnosti

Z-4.1.1.1. Dovršiti implementaciju koda za nedostajuće osnovne funkcionalnosti.

Padajući izbornik:

```
<select id="religion-select"></select>
```

```
select
  .selectAll("option")
  .data(religionList)
  .enter()
  .append("option")
  .attr("value", d => d)
  .text(d => d);
select.property("value", "Select religion");
```

```
export const religionList = ["Select religion", ...religionKeysOrdered.map(key =>
columnMapping[key])];
```

```
export const religionKeysOrdered = [
  "chrstprot",
  "chrstcat",
  "chrstorth",
  "chrstang",
  "chrstothr",
  "chrstgen",
  "judorth",
  // ...
```

```
export const columnMapping = {
  chrstprot: "Christianity Protestants",
  chrstcat: "Christianity Roman Catholics",
  chrstorth: "Christianity Eastern Orthodox",
  chrstang: "Christianity Anglican",
  chrstothr: "Christianity Others",
  chrstgen: "Christianity Total",
  judorth: "Judaism Orthodox",
  // ...
```

Izbornik za lakše snalazenje na stranici:

```
<div class="menu">
  <div>
    <a href="#section1">Country division</a>
```

```

</div>
<div>
  <a href="#section2">Regional division</a>
</div>
<div>
  <a href="#section3">Religion distribution</a>
</div>
<div>
  <a href="#" id="show-data-btn">Global data</a>
</div>
</div>

```

Gumb za resetiranje pogleda na kartu:

```
<button id="reset-btn-countries" style="display:none;">Back to World View</button>
```

```
const resetBtnCountries = d3.select("#reset-btn-countries");
```

```

resetBtnCountries.on("click", function(event, d) {
  resetBtn(zoomGroup, resetBtnCountries, infoBoxCountries);
  resetGraph(chartSvgCountries)
});

```

```

function resetBtn(zoomGroup, resetBtnCountries, infoBoxCountries) {
  zoomGroup.transition().duration(750).attr("transform", ""); // Reset transform
  resetBtnCountries.style("display", "none");
  infoBoxCountries.text("Select a country");
  selectedCountryCode = null;
  selectedCountryName = null;
}

```

Sekcije stranice zbog preglednosti:

```

<section id="section1">
  <h2 style="text-align: center;">Religion division by countries</h2>
  <div class="section1-container" style="display: flex; justify-content: space-between;">
    <div class="map-container">
      <div id="info-box-countries">Select a country</div>
      <svg id="map-countries" style="border: 2px solid #999;"></svg>
      <button id="reset-btn-countries" style="display:none;">Back to World View</button>
    </div>
    <div class="graph-container">
      <h3 style="text-align: center;">Country Data Visualization</h3>
      <svg id="graph-countries"></svg>
    </div>
  </div>
</section>

```

```

</section>

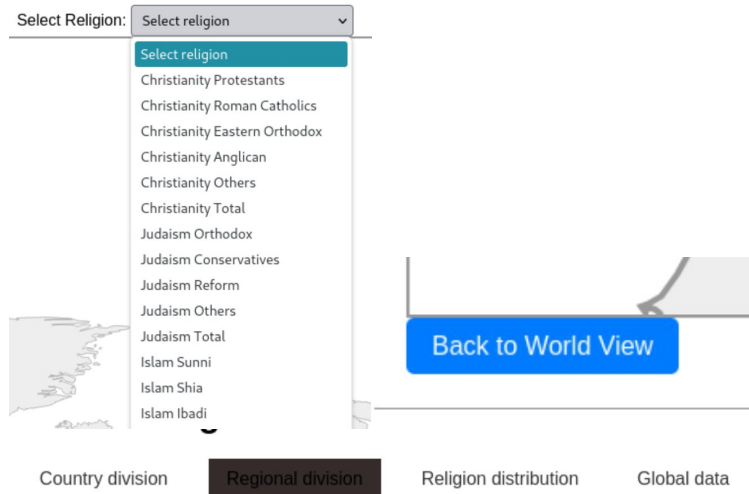
<section id="section2">
  <h2 style="text-align: center;">Religion division by continents</h2>
  <div class="section2-container" style="display: flex; justify-content: space-between;">
    <div class="graph-container graph-container-continents">
      <h3 style="text-align: center;">Continent Data Visualization</h3>
      <svg id="graph-continents"></svg>
    </div>
    <div class="map-container">
      <div id="info-box-continents">Select a continent</div>
      <svg id="map-continents" style="border: 2px solid #999;"></svg>
    </div>
  </div>

</section>

<section id="section3">
  <h2 style="text-align: center;">Religion distribution in world</h2>
  <div class="religion-dropdown-container" style="text-align: center; margin-top: 10px;">
    <label for="religion-select">Select Religion:</label>
    <select id="religion-select"></select>
  </div>
  <div class="section3-container" style="display: flex; justify-content: space-between;">
    <div class="map-container-distribution">
      <svg id="map-countries-distribution" style="border: 2px solid #999;"></svg>
    </div>
  </div>
</section>
</div>

```

- Z-4.1.2. Testirati osnovne funkcionalnosti i osigurati da su ispravne, tj. da funkcioniraju na očekivani način. Potrebno potvrditi slikom.



4.2. Implementacija osnovnog ponašanja

Z-4.2.1. Dovršiti implementaciju koda za nedostajuće osnovne interakcije.

Zoomiranje na klik

```
function processClickNationalEvent(event, d, infoBoxCountries, resetBtnCountries, zoomGroup, width, height, slider) {
  console.info("Clicked on country:", d.properties.name);
  const pathElement = d3.select(event.currentTarget);
  const bbox = pathElement.node().getBBox(); // Get the SVG bounding box in pixel space

  const { x, y, width: bboxWidth, height: bboxHeight } = bbox;
  if (bboxWidth === 0 || bboxHeight === 0) {
    return;
  }

  const scale = Math.min(8, 0.9 / Math.max(bboxWidth / width, bboxHeight / height));
  const centerX = x + bboxWidth / 2;
  const centerY = y + bboxHeight / 2;
```

```

const translate = [width / 2 - scale * centerX, height / 2 - scale * centerY];
// Apply transform to zoomGroup
zoomGroup.transition()
  .duration(750)
  .attr("transform", `translate(${translate}) scale(${scale})`)
const year = slider.node().value;
const countryName = d.properties.name || "Unknown Country";
const countryCode = getCountryCode(countryName, year);
const displayName = getDisplayName(countryName, year);
selectedCountryCode = countryCode;
selectedCountryName = displayName;
displayNationalData(countryCode, displayName);
drawGraph(countryCode, year);
resetBtnCountries.style("display", "block");
}

```

promjena godine prilikom koristenja slidera (plus ostale stvari)

```

slider.on("input", function () {
  const year = this.value;
  yearDisplay.text("Year: " + year); // ova linija koda
  if (selectedCountryCode && selectedCountryName) {
    drawGraph(selectedCountryCode, year);
    displayNationalData(selectedCountryCode, selectedCountryName, year);
  } else if (selectedReligionDistribution) {
    updateCountryColors(nationalData, selectedReligionDistribution, slider.node().value);
  } else if (selectedRegions) {
    displayRegionalData(selectedRegions, infoBoxContinents);
    drawRegionalGraph(selectedRegions, slider.node().value);
  } else {
    console.warn("NO COUNTRY SELECTED, SKIPPING GRAPH AND INFOBOX UPDATE");
  }
});

```

Promjena odabrane religije

```

select.on("change", function() {
  const religionKey = this.value;
  console.log("Religion selected:", religionKey);
  if (religionKey && religionKey !== "Select religion") {
    selectedReligionDistribution = religionKey;
    updateCountryColors(nationalData, religionKey, slider.node().value);
  } else {
    resetMapColors();
  }
});

```

Z-4.2.2. Testirati osnovno ponašanje i osigurati da je ispravno, tj. da funkcionira na očekivani način. Potrebno potvrditi slikom.

Religion distribution in world

Select Religion: Christianity Others ▼

Year: 1950



Year: 1945



III

4.3. Implementacija naprednih funkcionalnosti

Z-4.3.1. Identificirati napredne funkcionalnosti koje će biti implementirane.

Prikaz stupcastog graf-a za odabrane podatke

Bojanje karte s obzirom na gustocu religija

Prikaz podataka o drzavi u kutu, moguće scrollanje

Z-4.3.2. Implementirati napredne funkcionalnosti. Dokazati opisom primjera koda.

Prikaz stupcastog graf-a za odabrane podatke

```
export function drawBarChart({  
  svg,  
  data,  
  tooltip,  
  margin,  
  width,  
  height,  
  xLabel = "",
```

```

    yLabel = "Population"
  }) {
    const chartWidth = width - margin.left - margin.right;
    const chartHeight = height - margin.top - margin.bottom;

    // X and Y scales
    const x = d3.scaleBand()
      .domain(data.map(d => d.label))
      .range([margin.left, chartWidth + margin.left])
      .padding(0.1);

    const y = d3.scaleLinear()
      .domain([0, d3.max(data, d => d.value)])
      .range([chartHeight + margin.top, margin.top]);

    // Colors
    const colorScale = d3.scaleSequential()
      .domain([0, data.length - 1])
      .interpolator(t => d3.interpolateBlues(0.3 + t * 0.7));

    const colorMap = {};
    data.forEach((d, i) => {
      colorMap[d.label] = colorScale(i);
    });

    const chart = svg.append("g");

    chart.selectAll(".bar")
      .data(data)
      .join("rect")
      .attr("class", "bar")
      .attr("x", d => x(d.label))
      .attr("y", d => y(d.value))
      .attr("width", x.bandwidth())
      .attr("height", d => chartHeight + margin.top - y(d.value))
      .attr("fill", d => colorMap[d.label])
      .attr("data-original-fill", d => colorMap[d.label])
      .on("mouseover", function(event, d) {
        d3.select(this).attr("fill", "orange");
      });
  }

```

```

    tooltip
    .style("opacity", 1)
    .html(` ${d.label}: ${d.value.toLocaleString()} `)
    .style("left", (event.pageX + 10) + "px")
    .style("top", (event.pageY - 10) + "px");
  })
  .on("mouseout", function () {
    const original = d3.select(this).attr("data-original-fill");
    d3.select(this).attr("fill", original);
    tooltip.style("opacity", 0);
  });

// Axes
chart.append("g")
  .attr("class", "x-axis")
  .attr("transform", `translate(0, ${chartHeight + margin.top})`)
  .call(d3.axisBottom(x))
  .selectAll("text")
  .style("text-anchor", "end")
  .attr("dx", "-.8em")
  .attr("dy", ".15em")
  .attr("transform", "rotate(-45)");

chart.append("g")
  .attr("class", "y-axis")
  .attr("transform", `translate(${margin.left}, 0)`)
  .call(d3.axisLeft(y).tickFormat(d3.format(".2s")))
  .append("text")
  .attr("fill", "#000")
  .attr("transform", "rotate(-90)")
  .attr("y", 6)
  .attr("dy", "-3em")
  .attr("text-anchor", "end")
  .text(yLabel);
}

```

Bojanje karte s obzirom na gustocu religija

```

export function updateCountryColors(nationalData, selectedReligion, year) {
  const column = reverseColumnMapping[selectedReligion];
  if (!column) {
    console.error("INVALID RELIGION SELECTED:", selectedReligion);
    return;
  }
}

```



```

const yearStr = year.toString();
const dataForYear = nationalData.filter(d => d.year === yearStr);
if (!dataForYear || dataForYear.length === 0) {
  console.error("NO DATA FOR YEAR:", year);
  return;
}

const countryData = dataForYear.map(d => {
  const countryCode = d.name;
  const population = Number(d[column]) || 0;
  const mappedCountryCode = getCountryCode(countryCode, year);
  return { countryCode: mappedCountryCode, population };
}).filter(d => d.countryCode);

const populations = countryData.map(d => d.population).filter(p => p > 0);
const maxPopulation = d3.max(populations);
const minNonZeroPopulation = d3.min(populations.filter(p => p > 0)) || (maxPopulation * 0.01);
if (!maxPopulation || maxPopulation <= 0) {
  console.warn("NO VALID POPULATION DATA FOR:", selectedReligion, year);
  return;
}

const colorScale = d3.scaleLinear()
  .domain([minNonZeroPopulation, maxPopulation])
  .range(["#87CEEB", "#00008B"]);

const countryPaths = d3.select("#map-countries-distribution").selectAll("path");

countryPaths.each(function(d) {
  const path = d3.select(this);
  const countryName = d.properties.name || "Unknown Country";
  const countryCode = getCountryCode(countryName, year);
  const dataEntry = countryData.find(cd => cd.countryCode === countryCode);

  if (dataEntry) {
    if (dataEntry.population > 0) {
      path.style("fill", colorScale(dataEntry.population));
    }
  }
});

```

```

    } else {
      path.style("fill", "#ffffff");
    }
    path.on("mouseover", function(event) {
      const populationText = dataEntry.population > 0 ? formatNumber(dataEntry.population) :
"0";
      tooltip
        .style("opacity", 1)
        .html(` ${getDisplayNames(countryName, year)}<br>${selectedReligion}:
${populationText} `)
        .style("left", (event.pageX + 10) + "px")
        .style("top", (event.pageY - 10) + "px");
    })
    .on("mouseout", function() {
      tooltip.style("opacity", 0);
    });
  } else {
    path.style("fill", "#ffffff");
    path.on("mouseover", function(event) {
      tooltip
        .style("opacity", 1)
        .html(` ${getDisplayNames(countryName, year)}<br>${selectedReligion}: No data `)
        .style("left", (event.pageX + 10) + "px")
        .style("top", (event.pageY - 10) + "px");
    })
    .on("mouseout", function() {
      tooltip.style("opacity", 0);
    });
  }
});
}
}

```

Prikaz podataka o drzavi u kutu, moguće scrollanje

```

export function displayNationalData(countryCode, displayName) {
  if (!nationalData || nationalData.length === 0) {
    infoBox.text(` National data not available for ${displayName} `);
    return;
  }

  const year = slider.node().value;
  const dataForCountry = nationalData.filter(d => d.name === countryCode && d.year === year);

  if (dataForCountry.length === 0) {

```

```

    infoBox.text(`No data available for ${displayName} in ${year}`);
    return;
}

const row = dataForCountry[0];
let html = `<h4>${displayName} (${year})</h4>
    <table style="font-size: 12px;">
        <tr><th>Religion</th><th>Value</th></tr>`;

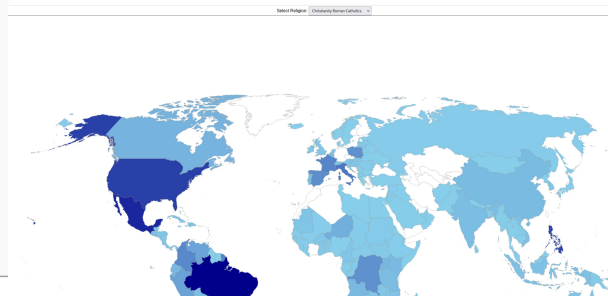
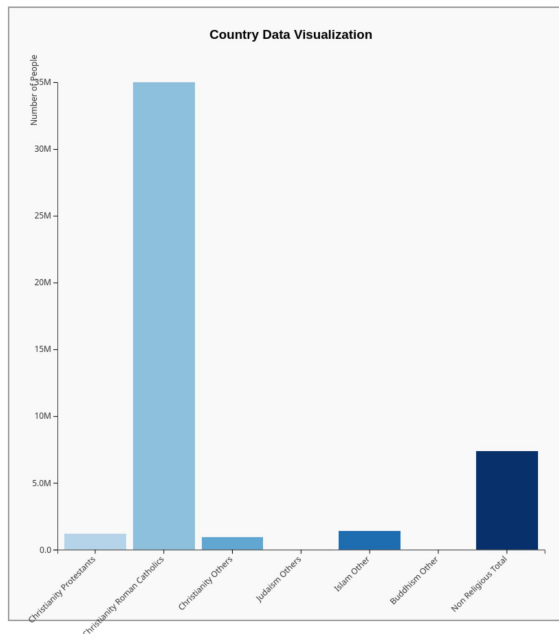
const entries = Object.entries(row).filter([(key)] => key !== "year" && key !== "state" && key !==
"name" && key !== "Version");
entries.forEach([(key, value)] => {
    const numValue = parseFloat(value.replace(/,/g, ""));
    if (!isNaN(numValue) && numValue !== 0) {
        const fullName = columnMapping[key] || key;
        const displayName = fullName.includes("Total") ? `<strong>${fullName}</strong>` : fullName;
        const formattedValue = formatNumber(value);
        html += `<tr><td>${displayName}</td><td>${formattedValue}</td></tr>`;
    }
});

html += '</table>';

infoBox.html(html);
}

```

Z-4.3.3. Testirati napredne funkcionalnosti i osigurati da su ispravne, tj. da funkcioniraju na očekivani način. Potrebno potvrditi slikom.



Portugal (2010)

Religion	Value
Christianity Protestants	85 140
Christianity Roman Catholics	8 780 066
Christianity Eastern Orthodox	63 855
Christianity Anglican	2 693
Christianity Others	191 565
Christianity Total	9 123 319
Judaism Others	3 192
Judaism Total	3 192
Islam Other	90 987
Islam Total	90 987
Buddhism Other	59 598
Buddhism Total	59 598
Hindu Total	6 386

4.4. Implementacija naprednog ponašanja

Z-4.4.1. Identificirati napredna ponašanja koje će biti implementirana.

Prelazak preko stupcastog grafa prikazuje podatke za taj stupac

Prelazak preko karte pokazuje podatke za tu drzavu

Prelazak preko karte mijenja ime drzave

Z-4.4.2. Implementirati napredna ponašanja. Dokazati opisom primjera koda.

Prelazak preko stupcastog grafa prikazuje podatke za taj stupac

```
const tooltip = d3.select("body")
  .append("div")
  .attr("class", "tooltip")
  .style("position", "absolute")
  .style("background", "#fff")
  .style("border", "1px solid #999")
  .style("padding", "5px")
  .style("border-radius", "3px")
  .style("pointer-events", "none")
  .style("opacity", 0);
```

```
.on("mouseover", function (event, d) {
  d3.select(this).attr("fill", "orange");
  tooltip
    .style("opacity", 1)
    .html(` ${d.label}: ${d.value.toLocaleString()} `)
    .style("left", (event.pageX + 10) + "px")
    .style("top", (event.pageY - 10) + "px");
})
.on("mouseout", function () {
  const original = d3.select(this).attr("data-original-fill");
  d3.select(this).attr("fill", original);
  tooltip.style("opacity", 0);
});
```

Prelazak preko karte pokazuje podatke za tu drzavu

```
const tooltip = d3.select("body")
  .append("div")
  .attr("class", "tooltip-distribution")
  .style("position", "absolute")
  .style("background", "#fff")
  .style("border", "1px solid #999")
  .style("padding", "5px")
  .style("border-radius", "3px")
  .style("pointer-events", "none")
  .style("opacity", 0)
  .style("font-family", "Arial, sans-serif")
  .style("font-size", "12px");
```

```
if (dataEntry) {
  if (dataEntry.population > 0) {
    path.style("fill", colorScale(dataEntry.population));
  } else {
    path.style("fill", "#ffffff");
  }
}
```

```

    }
    path.on("mouseover", function(event) {
        const populationText = dataEntry.population > 0 ? formatNumber(dataEntry.population) :
"0";
        tooltip
            .style("opacity", 1)
            .html(` ${getDisplayNames(countryName, year)}<br>${selectedReligion}:
${populationText}`)
            .style("left", (event.pageX + 10) + "px")
            .style("top", (event.pageY - 10) + "px");
        })
        .on("mouseout", function() {
            tooltip.style("opacity", 0);
        });
    } else {
        path.style("fill", "#ffffff");
        path.on("mouseover", function(event) {
            tooltip
                .style("opacity", 1)
                .html(` ${getDisplayNames(countryName, year)}<br>${selectedReligion}: No data `)
                .style("left", (event.pageX + 10) + "px")
                .style("top", (event.pageY - 10) + "px");
            })
            .on("mouseout", function() {
                tooltip.style("opacity", 0);
            });
        }
    }
});

```

Prelazak preko karte mijenja ime drzave

```

.on("mouseover", function(event, d) {
    infoBoxCountries.text(d.properties.name || "Unknown Country");
})
.on("mouseout", function(d) {
    if(!zoomedIn) {
        infoBoxCountries.text("Select a country");
    }
});

```

- Z-4.4.3. Testirati napredno ponašanje i osigurati da je ispravno, tj. da funkcionira na očekivani način. Potrebno potvrditi slikom.

